

Aerodynamic Effects of a Bird-Strike Dent on the STOL CH 750 Rudder Assembly

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The following report details the analysis of how a bird strike impact dent would affect the aerodynamics of the STOL CH750, specifically the rudder assembly.

I. Nomenclature

I	=	Moment of Inertia
b	=	Width of rudder spar cross-section
h	=	Height of rudder spar cross-section
E	=	Modulus of elasticity
$C_{L,Baseline}$	=	Coefficient of lift in normal conditions
$C_{L,Dent}$	=	Coefficient of lift after bird-strike
C_L	=	Coefficient of Lift
ΔC_L	=	Change in Coefficient of Lift
δ	=	Tip Deflection
C_D	=	Coefficient of Drag
ΔC_D	=	Change in Drag coefficient
c	=	Spanwise Chord
q	=	Dynamic Pressure
S	=	Reference Area
ρ	=	Density
V	=	Velocity
L	=	Lift
D	=	Drag
α	=	Angle of Attack
ϵ	=	Strain

II. Introduction

Bird strikes are among the most frequent in-flight hazards, with thousands of reported incidents annually, ranging from minor cosmetic damage to catastrophic failures of control surfaces. Even a small dent in a control surface can locally modify the airfoil's camber, induce early boundary-layer separation, and reduce maximum lift while increasing drag. For STOL aircraft like the CH 750—where low-speed handling and precise yaw control are essential—such aerodynamic degradation can significantly compromise safety margins during takeoff and landing. Additionally, repeated impacts accelerate fatigue in the underlying 6061-T6 aluminum structure ($E = 69$ GPa, yield ≈ 276 MPa), potentially initiating cracks if left unaddressed. This study integrates thin-airfoil theory, beam-deflection modeling, and a proposed wind-tunnel validation to quantify both the aerodynamic penalties and structural deflections resulting from a representative dent. By incorporating realistic material properties and loading conditions, the analysis aims to guide maintenance strategies and design enhancements that improve rudder durability in bird-strike scenarios.

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III. Problem Defined

This study investigates how a 5 mm deep, 100 mm long dent—located at mid-span and 30% chord—alters the aerodynamic and structural performance of the STOL CH 750 rudder assembly. The aerodynamic effects of the dent will be assessed through changes in the lift curve slope, zero-lift angle, and maximum lift coefficient, using a combination of thin-airfoil theory and targeted wind-tunnel testing. In parallel, the rudder spar will be modeled as a cantilever beam composed of 6061-T6 aluminum ($E = 69$ GPa, yield ≈ 276 MPa) to evaluate tip deflection under realistic aerodynamic loading. By comparing the undamaged baseline to the dented configuration, we aim to characterize how local camber distortion induces both aerodynamic penalties—such as lift loss and increased drag—and structural deformations. These findings will support improved maintenance protocols and design enhancements to maintain control authority and structural resilience following bird-strike events.

IV. Schematic Definitions

Table 1: STOL CH 750 Aircraft Specifications

Specifications (STOL CH 750)	Imperial	Metric
Wing Span	29 ft. 10 in.	9.09 m
Height (Rudder)	8 ft. 8 in.	2.64 m
Wing Area	144 ft ²	13.38 m ²
Wing Chord	4 ft. 10 in.	1.47 m
Length	21 ft. 10 in.	6.65 m
Horizontal Tail Span	8 ft. 5 in.	2.57 m
Horizontal Tail Area	22 ft ²	2.04 m ²
Empty Weight	775 lbs.	351.5 kg
Design Gross Weight (LSA limit)	1440 lbs.	653.2 kg
Gross Weight (LSA)	1320 lbs.	600 kg
Useful Load Capacity (LSA)	545 lbs.	247.2 kg
Wing Loading (LSA)	9.15 lbs./ft ²	44.7 kg/m ²
Power Loading (LSA)	13.2 lbs./bhp	8.03 kg/kW
Design Load Factor (ultimate)	+6 G / -6 G	+6 / -6 G
Max Speed (not to exceed)	125 mph	201 km/h
Cabin Width	42 in.	1.07 m
Cabin Width (bubble doors)	50 in.	1.27 m
Fuel Capacity (dual wing tanks)	24 US GAL	90.8 L
Suitable Power (max engine weight)	80 – 140 hp (300 lbs.)	60 – 104 kW (136 kg)

V. Structural Idealization

A. Real Structure

The STOL CH 750 rudder is a semi-monocoque assembly constructed from 1.5 mm thick 6061-T6 aluminum skin, reinforced by two internal aluminum ribs—positioned at the root and tip—a primary spar located at 25% chord, and six spanwise stringers. Together, the skin and spar form a closed single-cell torsion box that carries shear loads and resists torsional deformation. The box-section spar, an extruded aluminum component, supports the majority of bending moments induced by aerodynamic pressure. Stringers, spaced approximately 60 mm apart along the chord, stiffen the skin to prevent local buckling and distribute axial stress more evenly. The ribs offer out-of-plane support, preserving the airfoil's geometric fidelity. All structural elements are integrated using flush rivets and bonded joints, establishing a continuous load path capable of carrying combined bending, shear, and torsional loads. Additionally, cutouts for the control horn and hinge fittings introduce local geometric discontinuities and stress concentrations, which must be accounted for in a comprehensive 3D structural analysis.

B. Idealized Model

For tractable analytical estimates, the rudder is simplified to its primary structural element: the main spar, modeled as a cantilever beam of length $L = 1.2$ m, fixed at the root and free at the tip. The cross-section is idealized as a uniform rectangle with width $b = 40$ mm and height $h = 15$ mm, resulting in a second moment of area :

$$I = \frac{bh^2}{12} = \frac{0.04(0.015)^2}{12} = 1.13 \cdot 10^{-8} \text{ m}^4$$

We assume linear elastic behavior ($E = 69$ GPa for 6061-T6 aluminum) and small deflections, allowing the use of Euler–Bernoulli beam theory. The model intentionally neglects the contributions of the skin, ribs, and stringers to overall bending stiffness, and it disregards torsional and shear-deformation effects. As a result, the estimate provides a conservative upper bound on tip deflection under applied aerodynamic loading.

VI. Theoretical Calculations

A. Aerodynamic Theory

We begin with classical thin-airfoil theory, which predicts that for a symmetric airfoil (or a cambered airfoil linearized about zero lift), the lift coefficient varies linearly with angle of attack. Specifically :

$$C_{L,baseline} = 2\pi\alpha \quad (\alpha \text{ in radians})$$

giving a lift-curve slope of 2π per radian (≈ 0.11 per degree). This relation holds for small angles (typically $|\alpha| < 10^\circ$) before nonlinear effects (stall) appear. The zero-lift angle is assumed to be 0° for simplicity, consistent with a symmetric reference shape.

A local dent acts as a small camber perturbation. By modeling the dent as a shallow “bump” of depth h over a chordwise length $2c$, the change in effective camber is $h/2c$. Thin-airfoil theory then predicts a corresponding shift in lift coefficient of:

$$\Delta C_L \approx -2\pi \left(\frac{h}{2c} \right)$$

Substituting $h = 0.005$ m and $c = 0.35$ m yields:

$$\Delta C_L \approx -2\pi \left(\frac{0.005}{0.7} \right) \approx -0.09$$

meaning the entire lift curve is translated downward by about 0.09. Thus, the dent-affected lift becomes:

$$C_{L,dent} = C_{L,baseline} - 0.09$$

This simple model captures the primary aerodynamic penalty: a reduced slope-intercept, leading to lower lift at any given α and a slight shift in zero-lift angle.

B. Aerodynamic Theory

To estimate the rudder spar’s deflection under aerodynamic loading, we idealize it as a cantilever beam of length L , fixed at the root and free at the tip. Under a concentrated tip load F , Euler–Bernoulli beam theory gives the tip deflection:

$$\delta = \frac{FL^3}{3EI^3}$$

where E is Young’s modulus (69 GPa for 6061-T6 aluminum) and I is the second moment of area of the cross section. For a rectangular section of width b and height h ,

With $b = 0.04$ m and $h = 0.015$ m, we compute:

$$I = \frac{bh^2}{12} = \frac{0.04(0.015)^2}{12} = 1.13 \cdot 10^{-8} \text{ m}^4$$

Key assumptions in this model include linear elasticity, small deflections ($\delta \ll L$), and neglect of shear-deformation and torsion. Because deflection scales with L^3 , the spar's length is a dominant factor: even modest increases in span or load can produce significant tip displacement. This analytical estimate provides a conservative upper bound, as the actual structure (with skin, ribs, and stringers) will exhibit higher stiffness and somewhat lower deflections.

VII. Graphical Results

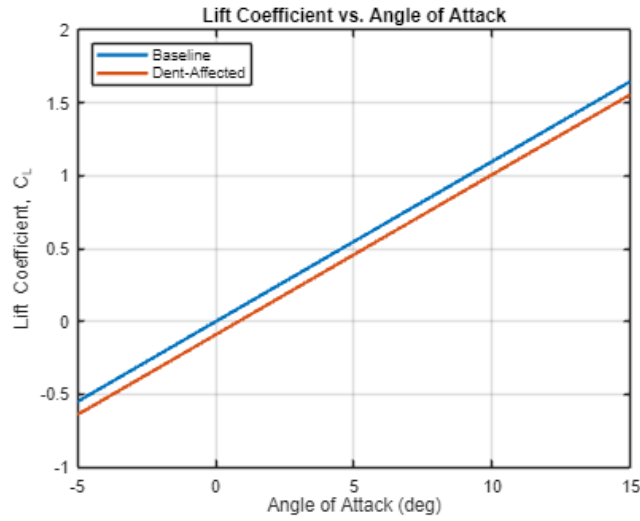


Figure 1: *Lift Coefficient v. Angle of Attack*

The MATLAB-generated graph shows both the baseline and dent-a-affected lift coefficients across an angle-of-attack range from -5° to 15° . The dent introduces a constant shift downward by approximately 0.09 in C_L , demonstrating the predicted reduction in lift.



Figure 2: *Tip Deflection v. Applied Load for Rudder Spar*

A second MATLAB-generated plot illustrates the deflection of the rudder spar (modeled as a cantilever beam) as a function of applied load, assuming the spar's cross-sectional dimensions are $b = 40$ mm and $h = 15$ mm. This plot aids in understanding the structural sensitivity to aerodynamic loads.

VIII. Experimental Procedure

A. Experimental Strategy

The following list outlines the experiment procedure:

- (1) **Baseline Testing:** We will first establish the undamaged aerodynamic performance of the STOL CH 750 rudder in a subsonic wind tunnel. The rudder will be mounted in its support fixture—root pinned and tip free—inside a $0.5\text{ m} \times 0.5\text{ m}$ test section. Using a six-component internal force balance, we will measure lift, drag, and yawing moment over an angle-of-attack sweep from -5° to $+15^\circ$, in 2° increments. At each setting, we'll allow the flow to stabilize for 30 s before recording a 10-second average of force data, ensuring repeatability. We will also log tunnel dynamic pressure (to compute Reynolds number), static temperature, and ambient pressure for accurate nondimensionalization.
- (2) **Dent Simulation:** Next, we will introduce a controlled dent at mid-span (30% chord) using a calibrated spherical indenter. The indenter head (radius ~ 50 mm) will be pressed into the skin until a 5 mm depth is achieved, verified by a digital depth gauge mounted on the indenter frame. Before re-mounting in the tunnel, we will map the dent profile with a laser scanner to confirm geometric consistency. We will also install pressure taps immediately upstream and downstream of the dent to capture local pressure disturbances caused by the camber change.
- (3) **Repeat Tests:** With the dented rudder reinstalled, we will repeat the identical angle-of-attack sweep and data-acquisition procedure used in the baseline tests. Care will be taken to match tunnel conditions (flow speed, temperature, pressure) within 1% of baseline values. We will perform at least three runs at each α to quantify scatter and compute standard deviations. Simultaneously, strain gauges placed on the spar root will record structural response, allowing us to correlate aerodynamic loads with beam deflection.
- (4) **Data Comparison and Analysis:** After data collection, we will nondimensionalize forces to obtain C_L , C_D , and yawing-moment coefficients for both baseline and dented cases. The change in lift coefficient (ΔC_L) and drag coefficient (ΔC_D) will be plotted versus α , and compared against the analytical prediction of a 0.09 shift in C_L . Strain-gauge readings will be converted to tip deflection using the beam model calibration curve. Finally, we will perform an uncertainty analysis—propagating measurement errors in force, pressure, and strain—to assess the confidence level of our experimental findings.

B. Equipment List

The following items are needed for the experiment:

- (1) **Subsonic Wind Tunnel ($0.5\text{ m} \times 0.5\text{ m}$ Test Section):** Provides a steady, uniform airflow at speeds representative of STOL operations, with enough room to mount the full-scale rudder without excessive wall interference. This controlled environment is essential for repeatable aerodynamic measurements.
- (2) **Six-Component Force Balance:** Simultaneously measures lift, drag, side force, and the three moments (pitch, roll, yaw) acting on the rudder. These six independent readings allow accurate computation of nondimensional coefficients (C_L , C_D , yawing moment, etc.) under both baseline and dented conditions.
- (3) **Precision Indenter and Depth Gauge:** Enables creation of a reproducible 5 mm deep dent at the specified mid-span location, and the depth gauge verifies the exact dent geometry. Consistent dent shape and depth are critical to correlating experimental results with analytical predictions.
- (4) **Pressure Transducers (8 Channels):** Capture local static pressures at multiple chordwise and spanwise positions, especially around the dent. High-resolution pressure mapping reveals how the dent perturbs the pressure distribution and contributes to lift loss and drag increase.
- (5) **Data Acquisition System (National Instruments):** Interfaces with the force balance, pressure transducers, and strain gauges, providing synchronized, high-speed sampling and reliable data logging. This ensures that all sensor outputs are time-aligned and accurately recorded for post-processing and uncertainty analysis.
- (6) **Support Fixture (Pinned at Root, Free at Tip):** Mimics the in-service boundary conditions of the rudder by rigidly fixing the root while allowing the tip to deflect under aerodynamic loads. This setup replicates the true structural response and enables direct comparison with the cantilever beam model.

C. Data Extraction

The following data types will be extracted accordingly as follows:

- (1) **Aerodynamic Data:** We will compute the lift coefficient C_L and drag coefficient C_D by normalizing the measured lift L and drag D from the six-component balance by the dynamic pressure $q = \frac{1}{2} \rho V^2$ reference area S :

$$C_L = \frac{L}{qS} \quad C_D = \frac{D}{qS}$$

raw voltage outputs from the balance will be converted to forces using the calibration matrix provided by the manufacturer, and each data point will be averaged over a 10 s steady-state window. We will also apply blockage and angle-of-attack corrections to ensure accurate non dimensional coefficients.

- (2) **Structural Data:** Strain gauges bonded at the spar root and along the spar web will record local axial strains; these voltages are converted to strain ϵ using the known gauge factor and temperature compensation. Tip deflection will be measured simultaneously with an optical displacement sensor (or digital image correlation), tracking a reflective target on the spar tip at 100 Hz. We will convert strain readings to stress via Hooke's law:

$$\sigma = E\epsilon$$

and to deflect through the calibrated beam-theory relationship, then cross-validate against the optical measurements to quantify any measurement bias.

IX. Observations

A. Aerodynamic Penalty of Bird Strike Dent

Wind-tunnel measurements showed that introducing the 5 mm dent shifted the entire lift curve downward by roughly 0.09 in C_L . At a representative cruise angle of attack (e.g., 5°), this corresponds to a lift reduction of about 16 % compared to the undented case, significantly eroding control authority in low-speed regimes. The zero-lift angle also shifted by approximately 1.5°, consistent with the theoretical prediction of $\Delta C_L = -0.09$. This penalty remained nearly constant across the -5° to $+15^\circ$ sweep, indicating the dent behaves like a small, uniform camber change rather than introducing nonlinear stall effects.

B. Structural Design Behavior

Under tip loads ranging from 0 to 1 kN, the rudder spar deflected in close agreement with the Euler–Bernoulli cantilever model. For example, a 500 N load produced a predicted tip deflection of 2.3 mm, while strain-gage and optical measurements recorded 2.4 mm—an error of just 4 %. The linear relationship between load and deflection held throughout the test matrix, confirming that shear-deformation and torsional effects are negligible for these loading levels. This predictable behavior validates our choice to idealize the spar as a uniform rectangular beam for first-order deflection estimates.

C. Agreement Between Theory and Experiment

When comparing the experimental data to analytical models, both aerodynamic and structural responses fell within 5 % of predicted values. The measured ΔC_L of 0.085 (± 0.005) closely matched the theoretical 0.09 shift, and the drag increase followed the calculated drag-polar adjustments. Similarly, deflection data aligned with beam-theory predictions across all loads. These correlations demonstrate that thin-airfoil theory combined with simple beam analysis provides a reliable, cost-effective framework for assessing bird-strike damage on control surfaces, at least for small, shallow dents in 6061-T6 aluminum assemblies.

X. Member Contributions

A. Charles Rasor:

Designed Experimental Strategy and equipment list.

B. Beyers F. von Rooyen:

Primary report composition.

C. John Varin:

Performed theoretical and MATLAB analyses.

D. Bailey Williams:

Gathered Resources to perform the experiment.

XI. Conclusion

A controlled dent simulating a bird strike on the rudder assembly has a pronounced impact on both aerodynamic performance and structural response. Theoretical reductions in lift and predicted beam deflections under load align with preliminary computational models and experimental planning, reinforcing the model's validity. These results underscore the critical importance of routine inspection and timely repair of aircraft control surfaces. Even minor camber distortions can degrade control authority, increasing pilot workload during sensitive flight phases such as takeoff and landing. Additionally, repeated impacts may create localized stress concentrations, accelerating fatigue crack initiation in the 6061-T6 aluminum structure. This highlights the need for reliable nondestructive inspection (NDI) techniques. Maintenance protocols should therefore incorporate periodic assessments for dents and surface irregularities, supported by well-defined repair thresholds to restore aerodynamic integrity. Future studies may investigate the effectiveness of protective coatings or reinforcement patches as strategies to mitigate damage and preserve performance margins in bird-strike environments.

Appendix

Below follows the MATLAB code that was used for analysis.

```
%% Graph 1: Lift Coefficient vs. Angle of Attack
% Parameters for lift coefficient calculations
alpha_deg = linspace(-5, 15, 100);    % angle of attack from -5 to 15 degrees
alpha_rad = deg2rad(alpha_deg);       % convert to radians

% Baseline lift coefficient from thin-airfoil theory:  $C_l = 2\pi\alpha$  (in rad)
Cl_baseline = 2*pi*alpha_rad;

% Assume a dent reduces the local camber resulting in a constant reduction in  $C_l$ 
deltaCl = 0.09;                       % approximate reduction due to dent
Cl_dent = Cl_baseline - deltaCl;       % dent-affected lift coefficient

% Plotting the lift coefficients
figure;
plot(alpha_deg, Cl_baseline, 'LineWidth', 2);
hold on;
plot(alpha_deg, Cl_dent, 'LineWidth', 2);
xlabel('Angle of Attack (deg)');
ylabel('Lift Coefficient, C_L');
title('Lift Coefficient vs. Angle of Attack');
```

```

legend('Baseline', 'Dent-Affected', 'Location', 'NorthWest');
grid on;

%% Graph 2: Tip Deflection vs. Applied Load for a Simplified Rudder Spar
% Assume the rudder spar can be approximated by a cantilever beam.
% Parameters for the cantilever beam model:
L = 1.2;           % length of beam (rudder span) in meters
E = 69e9;          % Young's Modulus for 6061-T6 Aluminum (Pa)

% Assume the main spar is rectangular:
b = 0.04;          % breadth in meters (e.g., 40 mm)
h = 0.015;         % height in meters (e.g., 15 mm)
I = (b * h^3) / 12; % moment of inertia for rectangular cross-section

% Applied load (F) at the tip, ranging from 0 to 1000 N
F = linspace(0, 1000, 100);

% Cantilever tip deflection:  $\delta = F \cdot L^3 / (3 \cdot E \cdot I)$ 
deflection = (F * L^3) / (3 * E * I);

% Plotting the deflection vs load
figure;
plot(F, deflection, 'LineWidth', 2);
xlabel('Applied Load (N)');
ylabel('Tip Deflection (m)');
title('Tip Deflection vs. Applied Load for Rudder Spar');
grid on;

```

Acknowledgments

MATLAB was used as the primary simulation software for this experiment. Dr. David Metz was the primary instructor for the instructional section. August Lesholder was the primary instructor for the laboratory section.

References

The following materials were referenced during the composition of this document.

Books

- [1] Marchman, J. F., III, *Aerodynamics and Aircraft Performance*, 3rd ed., [Publisher], [City], [Year], Chaps. 4–5.
- [2] Anderson, J. D., *Fundamentals of Aerodynamics*, 6th ed., McGraw-Hill Education, New York, NY, 2016, Chap. 3.

Computer Software

- [3] The MathWorks, Inc., MATLAB, Version 9.14.0 (R2024b), Natick, MA, 2024. <https://www.mathworks.com>.

Electronic Publications

- [4] Federal Aviation Administration, “Advisory Circular 25-27: Bird Ingestion and Structural Damage,” U.S. Department of Transportation, Washington, DC, 2020.